

Similarly, models can also be scaled and rotated accurately. This feature can be accessed by clicking on Object > Transform Object.

Modifying objects

Objects must be modified to be able to create anything meaningful. Modifying objects is complex in comparison to other 3D modelling software. Before the model is shaped, it must be converted into an editable object. To do this, first select the object, then click on Object > Convert to Triangular Mesh. Click OK. The tools are only accessible after double-clicking an object from the list on the right. The modelling is done in a different window altogether.

The window that opens has a similar layout to the main Art of Illusion program. There are four views and the toolbar is present on the right. This toolbar has many more tools to do with editing the selected object. Models can be modified and shaped mainly by moving the vertices, edges and planes around.

To switch to either mode, click on the Point, Edge or Face button on the bottom left of the window. Click on vertex, edge or face and move it around as you wish. You can have multiple selections by using the [Shift] key. Most modelling is done in this manner.

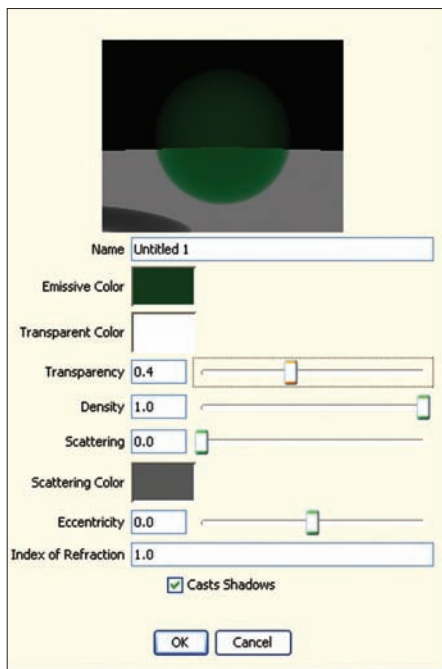
The tools provided are meant to help make modelling simpler, so you don't have to manually move each and every vertex or edge in place. Most of the tools are meant for moving elements in a symmetric manner. The other tools are for moving vertices along edges and also for bevelling and extruding planes. The extrude tool is particularly useful for starting with a simple object and extruding areas of it and adding detail to a model.

When all the modelling changes are completed, click on OK. Choose to save changes and the main window will be open again with the object updated.

Quick-modelling using boolean

One of the quickest ways that newcomers learn to model is by using the boolean feature. Although not always recommended, this method allows you to create simple objects by fusing two or more objects together or create complicated shapes. To use this feature, choose two more objects that are intersecting each other. Click on Tools > Boolean Modeling. Now choose the type of operation you want to use. A preview of the result is also displayed for guidance. For example, Union will add the two objects together and Intersection will choose the common piece of the two objects.

However, using the boolean feature in any modelling software isn't encouraged by some because it can spoil the flow of the wireframe and that creates complications while trying to map and texture the object.



A preview of the material used in Art of Illusion

Setting a material

With the model created, it's time to move to texturing. Without a texture, the model will still be lacking a big part of its look. The texture you want to use on the model can be anything from a photo or a texture created in Photoshop or any other image editor.

The material is only the basic look of the model. The texture is what gives it a realistic look. To set a material, first select the object, then right-click on it and click Set Material. They can also be set by choosing the material and texture from the drop-down menus in the properties pane on the right.

Materials and textures are resources that can be created and saved for use in later models and scenes. The textures and materials feature under the Scene menu. Images need to be imported to create textures. To do this, click on Scene > Images. Click on Load and select the images you want to import into Art of Illusion.

Adding lights and cameras

There is always a default light and camera when a scene is created. Additional lights are required to give a better effect for the scene. Lights and cameras are both simple to create. Tweaking the lights to create the ambience for the scene, on the other hand, is time consuming and requires patience.

Lights can be created by clicking on the Light icon on the toolbar. Click and drag a line in the viewport and let go to create a directional light. Double-click on the light in the Object list to change the type of light. The characteristics for the light such as the intensity and colour are visible in the properties tab.

Cameras can be added similarly. Double-clicking the camera lets you change settings for the field of view and depth of field. Special colour effects can be added by clicking on the Filters button. The newly created camera can be selected from the viewport's drop-down menu as well. During rendering, the animation or model can be shown through the new camera's view.

Rendering the model

The model that you see through your viewport is a low-quality version. The final version with all the models, textures and lights is only put together when you render it. To render a scene or object, click on the Scene menu and click on Render. Choose a resolution for the render, and set Antialiasing to Maximum. This will ensure a rendered model without jagged edges.

Once you have the settings set, then you can simply choose to quick-render it. To do this, click on Scene > Render Immediately.

If you are creating an animation, the rendering is also done using this same menu. Simply select Movie instead of Single Image, and set the time frame and the frame rate for the movie. Click OK. You will be prompted for the output file format and destination folder. You will need to use video-editing software to compile the image sequence into a video.

Customising Art of Illusion

Most of the features and interface settings that have been chosen may not suit you, especially, if you've been using a different 3D modelling software in the past. There are a few tweaks that you can do. Click on Edit > Preferences.

The viewports in Art of Illusion are set to show smooth models without any wireframe. If you prefer a wireframe where you can see all the edges and vertices, you can set that. You can also choose to see a textured version of the model right in the viewport instead of repeatedly rendering it.

You can modify shortcuts by clicking on the Shortcuts tab. Click on the shortcut you want to change, and click on Edit. If you know a little scripting, you can add some additional features linked to a shortcut key.

Art of Illusion is among the several free programs available for 3D modelling enthusiasts. Although its interface and workflow seem a bit messy up, once you get used to it, it's actually very usable.

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